Web Application Usage Instructions

1. Web Page initial

Our web application is located at https://majiapei.github.io/Landsat_VHR_RTSs/#. You can directly open the website in your browser without needing to log in. This page is for demonstration purposes only. No one is permitted to link to it from other websites.

Figure 1. The initial page.

The application provides zoom in/out, distance and area measurement tools (top-left controls in Figure 1), as well as a location search function. You can directly input latitude and longitude to switch the view to the target location.

2. Layer control

If the basemap fails to load correctly, open the layer control widget (Figure 2) in the lower-left corner of the map view and select a suitable basemap. You can also turn off the basemap and data layers.

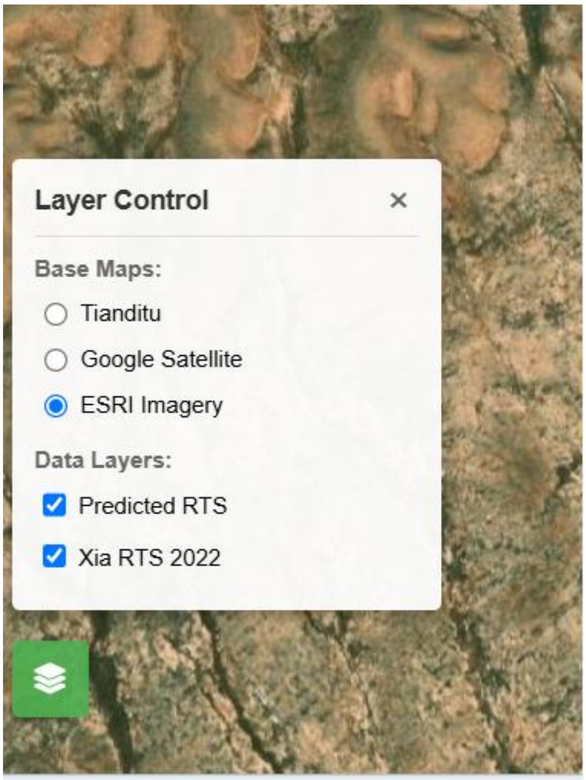

Figure 2. Layer control widget.

3. Analysis

To use the analysis functions, move your mouse to the right side of the page, and then click on the corresponding function. The application currently provides time series analysis and object detection features.

Figure 3. Activation of the analysis function.

3.1. Time Series Analysis

The time series analysis interface is shown in Figure 4

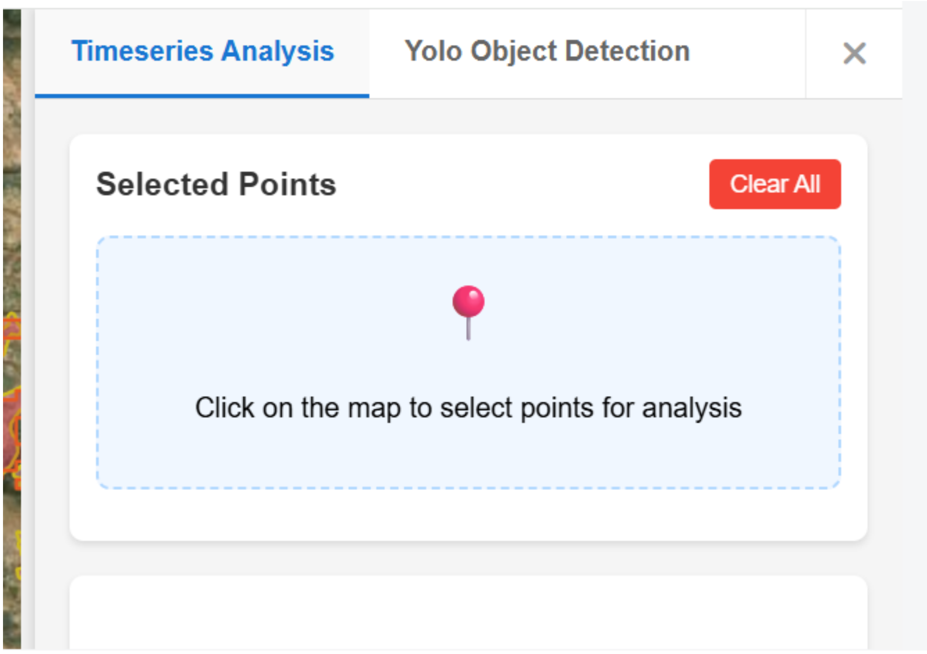

Figure 4. Time series analysis feature interface.

After clicking any point on the map, the application will plot the NDVI change for that point, the probability of disturbance as determined by the classification model, and the estimated start and end times of the disturbance provided by the time series segmentation model (Figure 5).

The feature also supports multi-point comparison, allowing users to, for instance, select a landslide area to analyze the differences in vegetation index changes at different elevations (Figure 6).

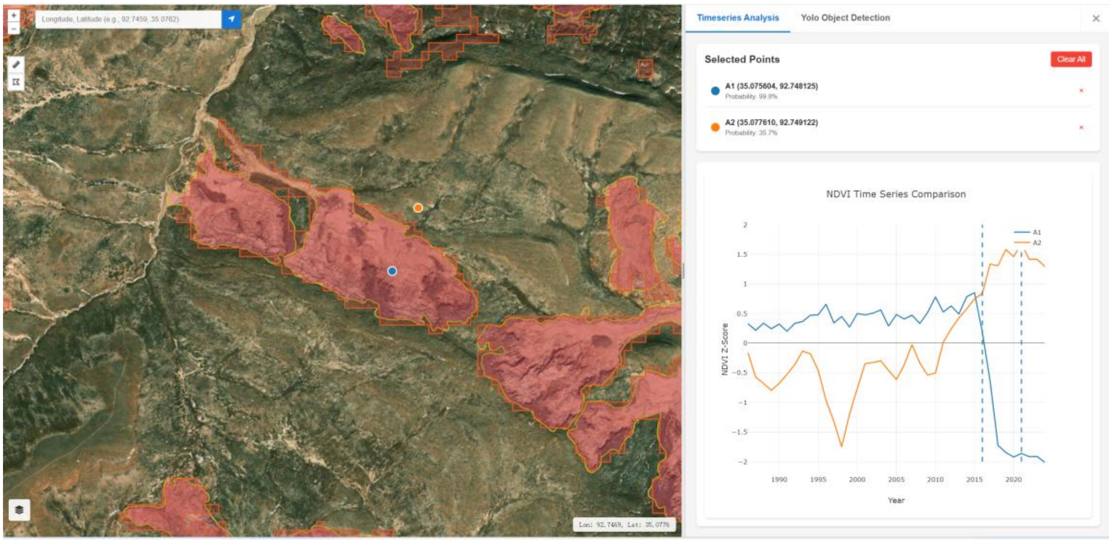

Figure 5. Demonstration of the time series classification and segmentation model results.

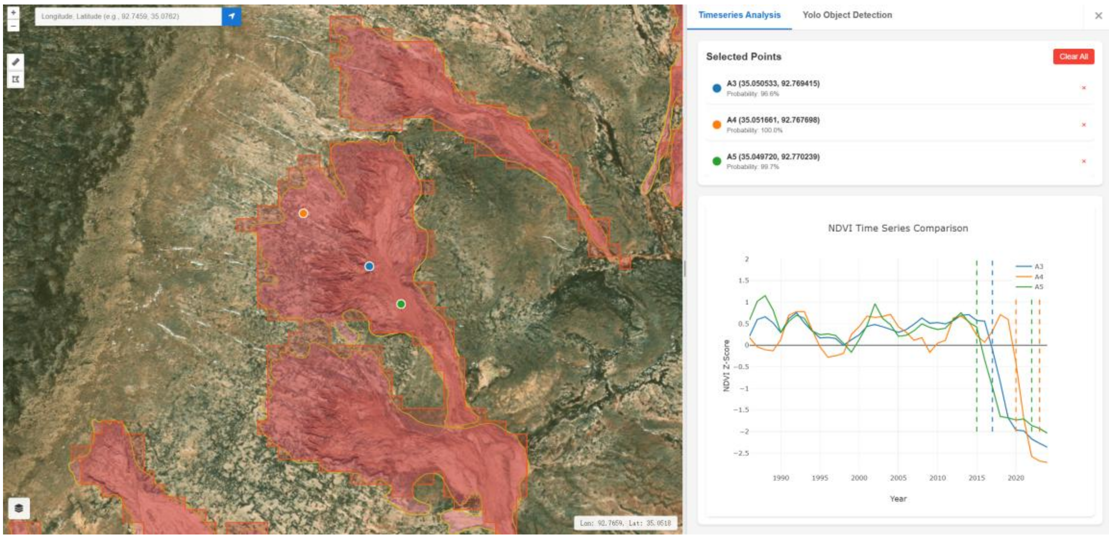

Figure 6. Comparative analysis of the timing of vegetation index changes at different elevations within the same RTS.

3.2. Object Detection Analysis

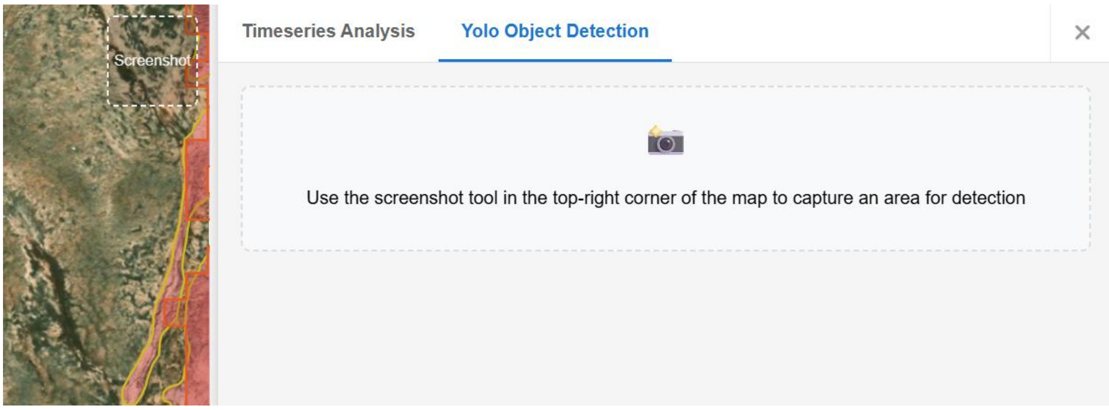

Figure 7. Activation of object detection analysis feature.

The object detection analysis feature takes a screenshot from the basemap and feeds it into a DL-based (Deep Learning) object detection model. The model then returns the results, which are displayed on the map.

After activating the feature, you must first click "screen shot" on the map page to enable the screenshot tool. You can capture multiple targets or a single target, but avoid selecting an area that is too large. Please try to ensure the screenshot area contains the entire RTS target.

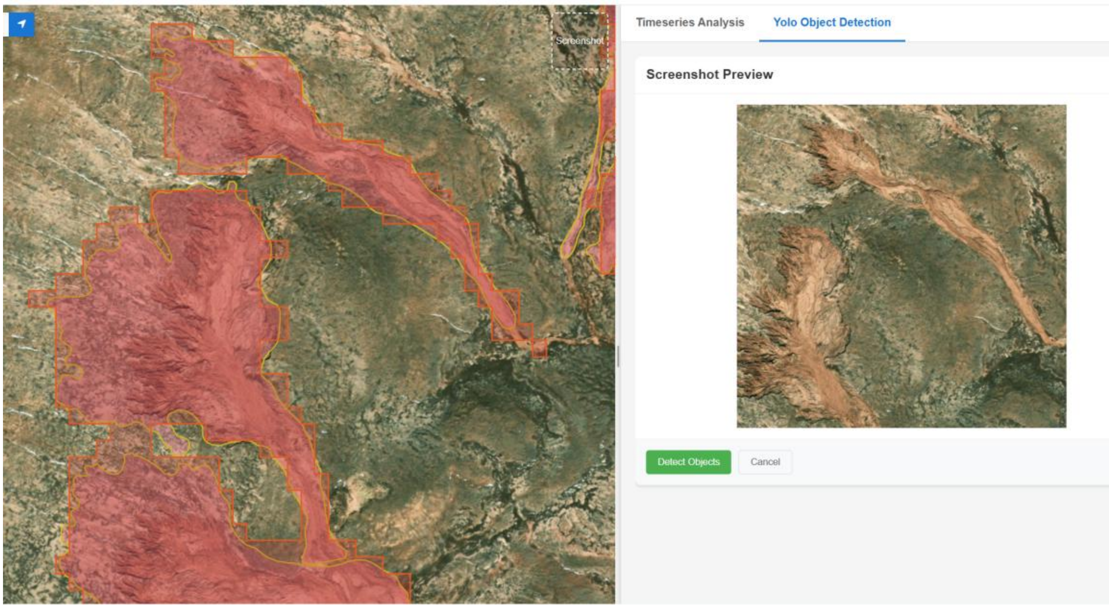

Figure 8. Screenshot feature demonstration.

Click the ‘Detect Objects’ button to start the detection

Figure 9. Object detection results display.